ECON-423/523-Labs-1-2

Sonan Memon

University of Oregon



Introduction

- I am a third year PhD Econ student; interested in international finance and monetary policy.
- My office hours are on Monday and Wednesday: 12 to 1 pm in PLC 418; email: smemon@uoregon.edu and course GitHub link is: github.

Installation of R, R Studio and Quarto.

- In this first lab, we will start with installing R, R Studio and working with Quarto.
- Download R, RStudio and Quarto.
- RStudio is the IDE; Quarto is for rendering presentations and papers. R markdown is a substitute for Quarto.

Common R Packages

Below, I have some common libraries loaded in R.

```
library(tinytex)

library(tidyverse)

library(dslabs)

library(dplyr)

library(ggplot2)


library(tibble)

library(modelsummary)
```

Common R Packages

- tidyverse: For data manipulation, visualization, and data import; dslabs: datasets to play with.
- tinytex: Tools for LaTeX/PDF compilation, particularly useful with Quarto.
- ggplot2: Data visualization using the Grammar of Graphics.
- tibble: modern, cleaner data-frame structure.

Common R Packages

- library(pacman)
- p_load(tidyverse, ggplot2, dplyr, tidyr, readr, stringr, lubridate)
- can load a bunch of packages separated by commas.
- dplyr: data manipulation and transformation, such as filtering, selecting, grouping, and summarizing; tidyr: reshaping and cleaning data, such as pivoting between wide and long formats.
- readr: importing data from CSV and other delimited text files; stringr — working with and cleaning text strings; lubridate — working with dates and times.

Basics of R/R Studio

- Why R? Many economists think it dominates STATA and R is open-source.
- Large online community for debugging.
- After Python, it is the second most common language for data science.
- More flexible than Stata; more functionality for metrics than Python.

Basics of R/R Studio

- Console: type code for trial without saving, bugs will display here.
- Render: Used for executing quarto; knitting is used for R Markdown.
- Environment: Shows data, variables and functions.
- Help: searching functions and syntax.
- .r (script file), .qmd or .rmd files are file types.

R-Markdown and Quarto

- In an R Markdown file,“‘{r, warning=FALSE, message=FALSE, eval=FALSE

means don't show warnings, don't show messages, and don't actually run this code.

- Quarto has largely replaced R-Markdown due to flexibility to include multiple languages but for this course, R Markdown should suffice.

Basic Functions

```
result = 5 + 2

sqrt(4)

# Objects do not need to be numbers:
memon <- "memon is the GE"

print(result)
print(memon)

class(result)
```

Basic Vector Operations

```
runs <- c(12, 65, 8, 55, 27, 15, 4, 5, 22, 30, 17,
          10, 40, 7, 29, 38, 5,
          23, 18, 20, 9, 25, 22, 0, 0)


summary(runs)

average <- mean(runs)
which.max(runs)

# sorting:
sort(runs, decreasing = TRUE)

# Position of Sorted Numbers:
order(runs, decreasing = TRUE)
```

Basic Vector Operations

- Computing statistics:

```
# element vise operation:

runs[c(1:20)]/20

average_first_20 <- mean(runs[c(1:20)])
average_first_5 <- mean(runs[c(1:5)])

# Percentiles:

quantile(runs, c(0.1, 0.5, 0.9, 0.99))

cat("Batting Average In First 20 Matches:",
    average_first_20, "\n")
```

Basic Data Wrangling

```
#setwd("C:/users/sonan/Documents")


tibble(
  name = paste0("Person", 1:1000),
  age = 20:1019
)


data <- read.csv("US_GDP_NX.csv")


head(data)

tail(data)
```

Basic Data Wrangling

```
data <- data.frame(data)



data$Quarter <- seq(from = as.Date("1947-01-01"),
                    to = as.Date("2020-12-01"),
                    by = 'quarter')

colnames(data) <- c("GDP", "NX", "index",
                    "Quarter")

# Create new variable with natural log
data$log_gdp <- log(data$GDP)

#write.csv(data, "output.csv", row.names = FALSE)
```

Equations

- Quarto allows for basic LaTeX syntax to produce equations.

$$y = X\beta + \varepsilon,$$

$$\hat{\beta}_{\text{OLS}} = (X'X)^{-1} X'y.$$

- $$y = \beta_0 + \beta_1 x + \beta_2 z + u$$

- $$\text{Bias}(\hat{\beta}_1) = \beta_2 \frac{\text{Cov}(x, z)}{\text{Var}(x)}$$

Binding 1A

```
country <- c("USA", "India", "Argentina", "Sudan")

WB_classification <- c("High Income",
                       "Lower Middle Income",
"Upper Middle Income", "Low Income")

rbind(country, WB_classification)

cbind(country, WB_classification)

cc <- cbind(country, WB_classification)
```

Binding 1B

```
chdi <- cbind(country, hdi_rank = c(17, 130,
                                    47, 176))

cbind(cc, chdi) # captures column name of latter

rbind(cc, chdi) # doesn't capture column names of
#latter.
```

Binding 2

```
country_class1 <- tribble(
  ~Country,      ~Income_Classification,
  "USA",         "High income",
  "India",       "Lower-middle income",
  "Argentina",   "Upper-middle income",
  "Sudan",       "Low income"
)

country_class2 <- tribble(
  ~Country,  ~Income_Classification,
  "USA",     "High income",
  "China",   "Upper-middle income",
  "Germany", "High income"
)
```

Binding 2

```
cc_tib <- as_tibble(cc)

chdi_tib <- as_tibble(chdi)

bind_cols(cc_tib, chdi_tib)

bind_rows(cc_tib, chdi_tib) # creates missing values

union(country_class1, country_class2)
# similar to bind rows but unions will remove
# duplicates.

intersect(country_class1, country_class2)

setdiff(country_class1, country_class2)
# unique of first
```

Mutating Joins 1

```
country_class3 <- tribble(~Country, ~HDI,
  "USA",          17,
  "India",        130,
  "Argentina",   47,
  "Ethiopia",     180
)


left_join(country_class1, country_class3,
          by = "Country")

left_join(country_class3, country_class1,
          by = "Country")
```

Mutating Joins 2

```
inner_join(country_class1, country_class3,
           by = "Country")

# above means merging.

full_join(country_class1, country_class3,
          by = "Country")
```

Filtering Joins

```
country_class3 <- tribble(
  ~Country,      ~HDI,
  "USA",          17,
  "India",        130,
  "Argentina",   47,
  "Ethiopia",     180
)


semi_join(country_class1, country_class3)
# keeps all rows of first where key matches
# with second.

anti_join(country_class1, country_class3)
# keeps rows of first which don't match
# with the second.
```

Outliers

```
data <- read.csv("output.csv")


stats <- summary(data$log_gdp)
stats

low <- 1.25
high <- 5.5

filtered_gdp <- filter(data, log_gdp > low,
                       log_gdp < high)
```

Regression

```
set.seed(1)

data <- tibble(x = runif(1000,0, 10),
               e = rnorm(1000,0,1),
               z = runif(1000, 0, 10),
               y = 10 + 1.5*x + 10*z + e)

model1 <- lm(data = data, y ~ x)

model2 <- lm(data = data, y ~ x + z)
```

Regression Table

```
O <- modelsummary(
  list("Regression X" = model1, "Regression X and
       Z" = model2),
  stars = TRUE,
  statistic = "std.error",
  output = "tinytable",
  title = "Results From Simulated Data",
  gof_omit = ".*"  # remove AIC, BIC, etc.
)
```

Regression Output

```
0
```

Table 1: Results From Simulated Data

	Regression X	Regression X and Z
(Intercept)	57.860***	9.866***
	(1.912)	(0.083)
x	1.895***	1.501***
	(0.331)	(0.011)
z		10.021***
		(0.011)

+ p < 0.1, * p < 0.05, ** p < 0.01, *** p < 0.001